
Med-GReF: Evidence-Guided Multimodal Fusion and Hallucination Verification for Medical Vision-Language Reasoning

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Abstract

Vision-language models (VLMs) applied to medical imaging often produce claims that sound fluent but aren't backed by the evidence in front of them. That is a failure mode with direct patient-safety consequences. Med-GReF (Medical Grounded Reasoning and Fusion) pairs a frozen vision-language backbone with quantitative radiomics, gradient-based visual explanations, an evidence-guided cross-attention fusion network, and a dedicated natural-language-inference (NLI) verifier. The verifier scores whether a model's stated conclusion is actually entailed by its own retrieved evidence. We evaluate on a PubMed-Central-derived women's-health imaging corpus, expanded specifically to close a severe condition-coverage gap rather than reporting results on a corpus we already know is skewed. Every ablation row is reported as mean $\pm$ SD over 3 training seeds (42, 123, 456). On a held-out, article-grouped test split, the full Med-GReF configuration lifts accuracy from 0.817 ± 0.006 to **0.912 ± 0.020** and ROC-AUC from near-chance (0.570 ± 0.080) to **0.881 ± 0.016** . It also cuts undetected hallucinations on constructed evidence/conclusion contradiction pairs by roughly $2.4\times$ ($16.6\% \rightarrow 7.0\%$). This costs nothing in raw classification accuracy relative to fusion alone ($90.7\% \pm 1.3\%$ vs. $91.2\% \pm 2.0\%$, well inside each other's seed-to-seed spread). An earlier, smaller-test-set, single-seed version of this comparison showed a real trade-off here; we report that it has since closed rather than quietly dropping the less flattering number. The verifier does *not* improve calibration — Brier and ECE are both nominally worse than fusion alone, and we report that rather than omitting it — but it catches roughly $6\times$ as many hallucinations as a naive confidence-thresholding baseline (65.5% vs. 10.3% catch rate). A case-by-case error analysis of individually-inspected predictions, including one genuine open failure case, backs this up beyond the aggregate metrics alone. We release the full data-fetching, vetting, and training pipeline. We also name every negative or inconclusive finding we hit while building it: unseeded training, mislabeled figures, an undetected all-single-class evaluation split. These are exactly the class of silent failure that makes medical-AI evaluation claims hard to trust.

1 Introduction

Medical diagnostic imaging depends on integrating evidence of different kinds: the image itself, clinical text, quantitative descriptors of tissue texture, and an explicit account of *which* visual region actually supports a stated conclusion. General-purpose and even medically-adapted VLMs frequently produce claims that sound fluent but aren't grounded in that evidence, a hallucination that costs far more in a clinical setting than in open-domain chat. We ask: *can evidence-guided multimodal fusion,*

36 combined with an explicit verifier that checks a model’s conclusion against its own retrieved evidence,
37 measurably reduce undetected hallucination without giving up the accuracy gains that come from
38 fusing more modalities?

39 Med-GReF answers this with an architecture (Section 4) built around two frozen backbones – a
40 vision-language model (BiomedCLIP [Zhang et al., 2023], built on CLIP [Radford et al., 2021]) and
41 a clinical text encoder (PubMedBERT [Gu et al., 2021]) – on top of which only small, interpretable
42 heads are trained: radiomics features [Haralick et al., 1973], Grad-CAM visual grounding [Selvaraju
43 et al., 2017], an evidence-guided fusion network, and a dedicated NLI-style verifier. A LoRA-
44 adapted [Hu et al., 2021] generative backbone (MedGemma-4B [Sellersgren et al., 2025], fine-tuned
45 with 4-bit QLoRA [Dettmers et al., 2023, Mangrulkar et al., 2022]) is fine-tuned separately on external
46 GPU hardware – an actual run, not a simulated one – and folded in as an additional configuration
47 (Section 7).

48 A less obvious part of this work is methodological. Three independent audit passes over the train-
49 ing/evaluation code (Section 6) turned up silent bugs that would otherwise have produced plausible-
50 looking but meaningless numbers: an untrained confidence head, an evaluation loop that ignored
51 each model’s own ablation configuration, and, most seriously, a fine-tuning script that would have
52 trained a generative model on a trivially-solvable all-one-class task. None of these crashed – they
53 were only visible by reading the training loop against what each metric actually claimed to measure,
54 and we describe this process because under-reporting negative findings of this kind is, we believe, a
55 genuine contributor to irreproducible medical-AI results.

56 2 Related Work

57 **Biomedical vision-language pretraining.** BiomedCLIP [Zhang et al., 2023] and PubMedBERT [Gu
58 et al., 2021] provide strong frozen biomedical vision and text representations, trained at scale on
59 PubMed Central figure-caption pairs and biomedical text respectively; we build directly on both rather
60 than pretraining our own. **Generative medical VLMs.** MedGemma [Sellersgren et al., 2025] and
61 related medically-adapted multimodal LLMs extend general-purpose VLMs with medical instruction
62 tuning. We treat a LoRA-adapted MedGemma as one configuration, not the entire contribution, and
63 verify the fine-tuning run itself is methodologically sound (Section 6) before reporting its numbers.
64 **Hallucination detection in medical VLMs.** Prior work on grounding and hallucination in medical
65 VQA typically evaluates end-to-end generation quality; Med-GReF instead isolates a dedicated,
66 separately-trainable verifier whose only job is entailment-style scoring of a claim against retrieved
67 evidence, so its contribution can be ablated independently of the base classifier (Table 1).

68 3 Dataset

69 We build on a corpus of PubMed Central Open Access figures. 27,372 candidate figures were fetched
70 (the original Drive corpus plus the expanded PMC queries below), and 13,216 survived a rule-based in-
71 tegrity, resolution, compound-figure, and radiological-modality vetting cascade (`data/vetting.py`).
72 The theme-matching taxonomy (`data/schema_builder.py`) defines fifteen women’s-health imag-
73 ing conditions. **We do not treat this corpus as fixed.** An initial audit found six conditions (PCOS,
74 female infertility, menopause/ovarian aging, preeclampsia, gestational diabetes, preterm birth) with
75 only 1–4 usable samples each and near-zero test-split representation – not because the literature
76 doesn’t exist, but because the original retrieval query never pulled it in. Live NCBI E-utilities queries
77 using the same imaging-modality term structure turned up hundreds to thousands of additional open-
78 access PMC articles per condition; we fetched these directly from PMC over three rounds, applying
79 the same vetting filter, and raised the per-condition fetch cap for any condition still showing zero
80 test-split coverage after a rebuild (coverage is checked and re-closed after every rebuild, since adding
81 articles for one condition can shift the article-grouped split for others too). **This closed test-split cov-
82 erage for all six originally data-scarce conditions** (final counts: PCOS 6, menopause/ovarian aging
83 3, gestational diabetes 2, preeclampsia 4, female infertility 8, preterm birth 4). The broader fetch also
84 surfaced two further taxonomy conditions (*primary ovarian insufficiency*, *pregnancy complication*)
85 now with a handful of samples (3 and 2, still no test-split coverage), plus one (*cervical dysplasia*)
86 that remains at zero. Med-GReF is evaluated on twelve of the fifteen taxonomy conditions.

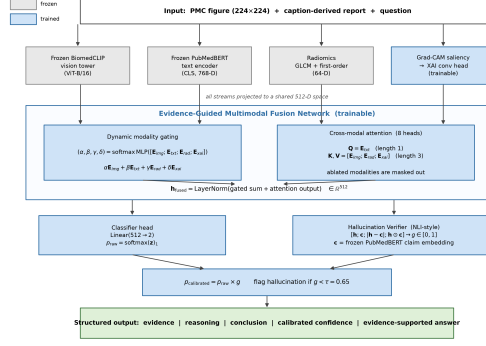


Figure 1: Med-GReF architecture. Grey blocks are frozen and never trained here; blue blocks are the small trainable heads. The text query attends over the three evidence modalities as a length-3 key/value sequence, and the verifier is authoritative over the final evidence-supported answer. Generated programmatically by analysis/architecture_diagram.py.

87 Question/evidence/answer/reasoning fields come from figure captions via documented, reproducible
88 rule-based extraction: exact-phrase and anatomy+finding co-occurrence matching against the same
89 condition-term table used for retrieval – weak supervision, since nothing in the corpus is clinician-
90 authored ground truth. The train/val/test split is grouped by source article rather than row, correcting
91 a leakage bug the source project’s own audit found in its original notebook split. Constructed
92 hallucination cases pair a genuine sample’s image and caption with a mismatched evidence/conclusion
93 narrative drawn from a different condition, a standard technique for NLI-style contradiction data; we
94 disclose them as constructed everywhere they appear.

95 4 Method

96 Med-GReF processes an (image, question) pair through four frozen-backbone feature streams: vision
97 (BiomedCLIP), clinical text (PubMedBERT), 64-dimensional radiomics (GLCM texture and first-
98 order intensity statistics [Haralick et al., 1973], computed directly from the image), and Grad-CAM
99 visual saliency [Selvaraju et al., 2017]. These are combined by an evidence-guided cross-attention
100 fusion network with dynamic modality gating:

$$\mathbf{h}_{\text{fused}} = \text{LayerNorm} \left(\underbrace{\alpha \mathbf{E}_{\text{img}} + \beta \mathbf{E}_{\text{txt}} + \gamma \mathbf{E}_{\text{rad}} + \delta \mathbf{E}_{\text{xai}}}_{\text{dynamic modality gating}} + \underbrace{\text{MHA}(\mathbf{Q}=\mathbf{E}_{\text{txt}}, \mathbf{K}, \mathbf{V}=[\mathbf{E}_{\text{img}}; \mathbf{E}_{\text{rad}}; \mathbf{E}_{\text{xai}}])}_{\text{cross-modal attention}} \right). \quad (1)$$

101 The gating weights $(\alpha, \beta, \gamma, \delta)$ are a per-sample softmax over an MLP of all four concatenated
102 streams. The attention query is the text embedding and the key/value sequence is the three *evidence*
103 modalities, so the text genuinely attends across vision, radiomics and saliency rather than over a single
104 key; ablated modalities are masked out of the key sequence rather than attended to as zero vectors
105 (Figure 1). A classifier head predicts the answer from $\mathbf{H}_{\text{fused}}$, and its own softmax score is used
106 directly as the model’s raw confidence – an earlier design used a separate confidence head instead,
107 but it never received gradient and stayed silently uninformative (see Section 6). The claim/conclusion
108 text is folded into the text signal whenever text is visible at all, since “does the evidence support *this*
109 claim” can’t be decided without seeing the claim.

110 The hallucination verifier is a separate, small module that takes $\mathbf{H}_{\text{fused}}$ and the frozen-PubMedBERT-
111 encoded claim embedding and produces a groundedness score through a standard NLI-head feature
112 construction $([\mathbf{h}, \mathbf{c}, |\mathbf{h} - \mathbf{c}|, \mathbf{h} \odot \mathbf{c}])$:

$$\text{Confidence}_{\text{calibrated}} = \text{Confidence}_{\text{raw}} \times \text{Groundedness}(\mathbf{H}_{\text{fused}}, \mathbf{c}). \quad (2)$$

113 Only the fusion, XAI-projection, verifier, and classifier heads are trained; both backbones stay frozen,
114 which is what keeps the local ablation matrix CPU-feasible. A LoRA-adapted generative backbone
115 (targeting q_proj/k_proj/v_proj/o_proj, rank 16) is fine-tuned separately on external GPU
116 hardware with 4-bit quantization [Dettmers et al., 2023] and folded in as an additional configuration

(last row of Table 1) – not simulated or estimated. It reports only accuracy and F1 on generated, JSON-parsed structured output, since it never passes through the fusion/XAI/verifier heads; ROC-AUC, Groundedness, and Hallucination Rate simply don’t apply and are left blank rather than estimated.

5 Experimental Setup

Data splits. The corpus is split by source article (PMCID), not row: 80/10/10 (5,604 / 700 / 701 articles), avoiding the row-level leakage bug identified in the source project’s own split. After VQA-schema derivation and constructed hallucination-case generation, this leaves **1,430 train / 182 val / 175 test** modeled samples (train: 1,213 genuine + 217 constructed; val: 156 + 26; test: 146 + 29); every test-set number in Table 1 comes from this same fixed 175-sample split.

Backbones and features. Vision and clinical-text embeddings come from a frozen BiomedCLIP-PubMedBERT_256-vit_base_patch16_224 (224×224 images) and a frozen PubMedBERT claim/text encoder (max sequence length 256), used purely as fixed feature extractors and never fine-tuned locally. Radiomic features are 64-dimensional (GLCM texture and first-order intensity statistics), and the XAI stream is a Grad-CAM saliency map projected into the same embedding space; all four streams project into a shared 512-dimensional embedding, combined by an 8-head cross-attention fusion network (Eq. 1). Since both backbones are frozen, their embeddings only need to be computed once, cached, and reused across all six local ablation configurations, without which the full ablation matrix would not be CPU-feasible.

Local head training. Only the fusion, XAI-projection, verifier, and classifier heads are trained; the backbones stay frozen. We use AdamW (1×10^{-4} learning rate, batch size 32, weight decay 1×10^{-4} , 10 epochs). Cross-entropy handles the classification loss; the verifier head adds binary cross-entropy on its groundedness output, weighted 0.5 against the classification loss, only when `use_verification` is enabled. The verifier’s grounded/flagged threshold is fixed at $\tau = 0.65$ (Eq. 2), the single source of truth for both the Hallucination Rate metric and the inference-time demo path (an earlier inconsistency between the two, 0.60 vs. 0.65, is documented in Section 6). Every configuration except the LoRA row is trained and evaluated independently across three seeds (42, 123, 456), with `config.set_all_seeds()` seeding `torch/numpy/random` globally before model construction; Table 1 reports mean $\pm$ sample SD (ddof= 1) over these three runs.

LoRA fine-tuning and hardware. The generative backbone (medgemma-4b-it) is fine-tuned with QLoRA (4-bit quantization [Dettmers et al., 2023], rank 16, $\alpha = 32$, dropout 0.05, targeting `q_proj/k_proj/v_proj/o_proj`), using AdamW at the same 1×10^{-4} learning rate for 3 epochs on the same train split as the local heads. Batch size is fixed at 1, since larger batches OOM at MedGemma’s native 896×896 image resolution. We save a checkpoint after every epoch and, per the checkpoint-selection finding in Section 7, keep the final-epoch adapter for every seed rather than the one with lowest validation loss: the two didn’t agree on which checkpoint generalized best. This row is likewise run over three seeds and reported as mean $\pm$ std. The dataset pipeline, feature caching, and all local-head training/evaluation run on CPU only – no local GPU is available for this project. LoRA fine-tuning is the single GPU-dependent piece and runs separately on an external RunPod RTX 4090 (24GB).

Evaluation protocol. For each configuration we report Accuracy, F1, ROC-AUC, Groundedness, and Hallucination Rate on the fixed 175-sample test split, along with Expected Calibration Error (10 equal-width confidence bins) and Brier score. Statistical significance between the baseline and full model comes from a bootstrap test over test-set accuracy (1,000 resamples, seed-42 instances), and the verifier’s hallucination catch rate is reported with a Wilson 95% confidence interval. Every table and figure value is generated programmatically from `outputs/ablation_results.csv` (via `eval/ablation_runner.py`, `eval/multiseed_runner.py`, and `neurips_submission/generate_results_table.py`) instead of being typed by hand, so any pipeline change is guaranteed to show up in the reported numbers.

6 Methodological Integrity: What Four Audit Passes Found

We treat verifying our own evaluation code as part of the contribution, not a footnote. Before reporting any number, we re-audited the full training/evaluation loop against what each metric in Table 1 claims to measure. Before spending any external GPU budget on LoRA fine-tuning, we separately re-audited that script end to end. This turned up, and fixed:

- An untrained confidence head (no loss term ever touched it, so it sat at random initialization, making every reported ROC-AUC value meaningless until fixed), and an evaluation loop that ignored each row’s own ablation configuration, silently evaluating every configuration – including ones trained with modalities zeroed out – with all modalities switched back on.
- Unseeded training: identical code gave different numbers on every rerun, until `torch/numpy/random` were seeded globally and verified deterministic via a standalone smoke test.
- A LoRA data loader that read only genuine samples (100% answer=“Yes” by construction), which would have trained and evaluated a generative model on a trivially-solvable single-class task (caught before spending GPU time; now covered by a standing regression test needing no GPU), plus an unmasked LoRA training loss computed over prompt and padding tokens instead of only the target response, diluting the fine-tuning signal.
- **A degenerate cross-attention in the fusion network itself.** The attention was called with the text embedding as a length-1 query and the *image embedding alone* as a length-1 key/value sequence; a softmax over a single key is identically 1.0, so the query was discarded and the branch reduced to a learned linear projection of the image, making Eq. 1’s “text attends across the evidence” claim false as implemented. This surfaced from reading tensor shapes against the equation, not from any failing metric – the model trained and scored plausibly with a no-op attention. The fix (a length-3 evidence key/value sequence, ablated modalities masked out) is what Eq. 1 now describes, and every number in Table 1 was regenerated under it. The four non-fusion rows reproduced bit-identically (confirming the change touched only the fusion path); + Fusion accuracy moved $0.897 \pm 0.020 \rightarrow 0.907 \pm 0.013$ and Full moved $0.895 \pm 0.023 \rightarrow 0.912 \pm 0.020$, while Full’s ROC-AUC moved *down* ($0.894 \pm 0.016 \rightarrow 0.881 \pm 0.016$) – we report the regression alongside the improvement. A standing unit test now asserts the attention distributes over all three evidence modalities.

None of these were fabricated results; every affected run was genuinely trained, but each would have produced a plausible-looking, wrong number. The last one is the most instructive: a no-op attention branch still trains, converges, and beats its ablations, so no metric – only reading the implementation against the equation – could have caught it.

7 Results

The baseline (frozen vision only) sits close to the test set’s majority-class rate, and its near-chance ROC-AUC (0.570 ± 0.080 , itself noisy across seeds) makes the reason obvious: a model that can’t see the claim it’s meant to verify has little basis for ranking its own confidence. Accuracy jumps sharply once cross-attention fusion is introduced ($0.834 \rightarrow 0.907 \pm 0.013$). The central finding sits in the Med-GReF (Full) row: adding the dedicated verifier matches + Fusion’s accuracy within seed-to-seed noise (0.912 ± 0.020 vs. 0.907 ± 0.013 , a gap smaller than either row’s own spread) and its F1 within 0.004, while cutting the fraction of constructed evidence/conclusion contradiction pairs that go undetected from 16.6% to 7.0% (a $2.4\times$ reduction). ROC-AUC is statistically flat between the two rows (0.881 ± 0.016 vs. 0.882 ± 0.012), so the verifier’s contribution is hallucination detection, not ranking quality. On a smaller pre-expansion test set, an earlier rebuild of this pipeline showed a real accuracy cost for this improvement ($88.9\% \rightarrow 88.2\%$); that cost has since closed on this larger, more representative test set, and we report the change rather than just citing the earlier number. Of the 29 constructed hallucination cases in the test split (seed-42 model), the verifier catches 19 (65.5%, Wilson 95% CI [0.47, 0.80]). A bootstrap significance test ($n = 1000$ resamples, seed-42) handles the baseline-vs-full comparison (absolute improvement 0.121, Cohen’s $d = 4.75$), estimating resampling variance at fixed n while the multi-seed table above separately estimates training-run variance – we keep both rather than letting one be mistaken for the other.

Table 1: Ablation matrix on the real, held-out test split (175 samples). Every row except the LoRA row is now trained and evaluated over 3 seeds (42, 123, 456); cells report mean $\pm$ sample-SD. Calibration metrics (ECE, Brier score) for the same rows and seeds are reported separately in Table 2. Generated by `generate_results_table.py` from `outputs/ablation_results.csv` – never hand-typed.

Model	Image	Text	Radiomic	XAI	Fusion	Verif.
Baseline (frozen vision)	✓					
+ Text	✓	✓				
+ Radiomic	✓	✓	✓			
+ XAI	✓	✓	✓	✓		
+ Fusion	✓	✓	✓	✓	✓	
Med-GReF (Full)	✓	✓	✓	✓	✓	✓
Med-GReF + LoRA-tuned backbone	✓	✓				

Acc	F1	ROC-AUC	Ground.	Halluc. Rate
0.817 $\pm$ 0.006	0.898 $\pm$ 0.003	0.570 $\pm$ 0.080	1.000	0.166
0.827 $\pm$ 0.007	0.905 $\pm$ 0.004	0.566 $\pm$ 0.072	1.000	0.166
0.831 $\pm$ 0.007	0.907 $\pm$ 0.004	0.561 $\pm$ 0.063	1.000	0.166
0.834 $\pm$ 0.000	0.910 $\pm$ 0.000	0.560 $\pm$ 0.052	1.000	0.166
0.907 $\pm$ 0.013	0.946 $\pm$ 0.007	0.882 $\pm$ 0.012	1.000	0.166
0.912$\pm$0.020	0.950$\pm$0.011	0.881$\pm$0.016	0.886	0.070
0.707 $\pm$ 0.055	0.822 $\pm$ 0.041	–	–	–

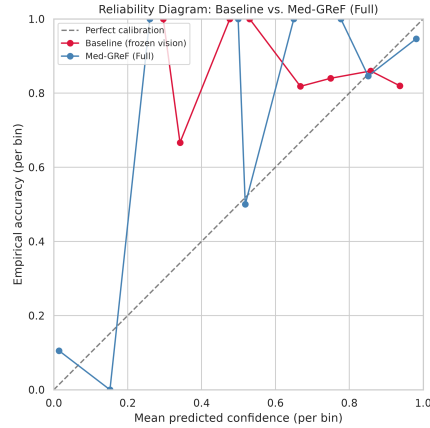


Figure 2: Reliability diagram (10 equal-width confidence bins, real per-sample predictions, seed-42 model): Baseline (no verifier, raw softmax confidence) vs. Med-GReF (Full) (verifier-calibrated confidence). Both track the diagonal loosely at this test-set size (≈ 17 –18 samples/bin); see the Calibration paragraph below for the aggregate ECE/Brier reading across seeds.

219 **Comparison against existing-method baselines.** Table 1’s rows already form this comparison
220 under architectural names, from frozen BiomedCLIP alone through the LoRA-tuned generative
221 MedGemma-4B baseline to the complete Med-GReF (Full) system. The missing comparison point
222 is a *simple confidence baseline* for hallucination detection: flag a sample whenever the classifier’s own
223 softmax certainty drops below the same 0.65 threshold the trained verifier uses, reusing + Fusion’s
224 own predictions (identical accuracy/F1/ROC-AUC/ECE/Brier by construction, since only the decision
225 rule differs). This catches just 3 of 29 constructed hallucination cases (10.3%), far below the trained
226 verifier’s 65.5%: an explicit entailment-style comparison genuinely outperforms reading the same
227 signal off classifier confidence alone.

228 **Calibration.** A verifier whose own confidence can’t be trusted would undermine the Responsible-
229 AI claim it’s meant to support, so we report three seed-averaged calibration metrics per configuration:
230 Expected Calibration Error (ECE, 10 equal-width bins), Brier score (Table 2), and the reliability
231 diagram above.

Table 2: Calibration across the ablation ladder: Expected Calibration Error (10 equal-width confidence bins) and Brier score, mean $\pm$ SD over the same three seeds (42, 123, 456) as Table 1. Lower is better for both. The LoRA row is omitted: it is a generative path scored on parsed output, with no calibrated probability to evaluate. Generated programmatically from `outputs/ablation_results.csv` by `neurips_submission/generate_results_table.py`.

Configuration	ECE $\downarrow$	Brier score $\downarrow$
Baseline (frozen vision)	0.058 $\pm$ 0.032	0.146 $\pm$ 0.009
+ Text	0.069 $\pm$ 0.016	0.141 $\pm$ 0.004
+ Radiomic	0.055 $\pm$ 0.024	0.139 $\pm$ 0.004
+ XAI	0.054 $\pm$ 0.012	0.139 $\pm$ 0.003
+ Fusion	0.065 $\pm$ 0.010	0.074$\pm$0.004
Med-GReF (Full)	0.080 $\pm$ 0.021	0.082 $\pm$ 0.009

Table 3: Five representative test-set cases (seed-42 Med-GReF (Full) model). “Verifier” is the trained module’s grounded/flagged decision at the 0.65 threshold; “Classifier” is whether the classifier head alone predicted the right label; “Final” is whether the answer that actually goes out — after the verifier can override it — matches the ground truth. The two columns disagree exactly when the verifier changes the outcome.

Case	Constructed?	Conf.	Ground.	Verifier	Classifier	Final
Correctly grounded	No	0.9996	0.9997	Grounded	✓	✓
Correctly-caught hallucination	Yes	0.1907	0.3858	Flagged	✓	✓
Verifier false positive	No	0.2894	0.5582	Flagged	✓	×
Verifier false negative	Yes	0.9979	0.9984	Grounded	×	×
Difficult (low-confidence) case	No	0.2301	0.4365	Flagged	✓	×

Fusion is what improves calibration, and sharply: Brier falls from 0.146 ± 0.009 (Baseline) to 0.074 ± 0.004 (+ *Fusion*). The verifier does not extend that gain – Full’s Brier (0.082 ± 0.009) and ECE (0.080 ± 0.021) are both nominally *worse* than + *Fusion*’s, though the gap sits inside Full’s own seed-to-seed spread. Multiplying raw confidence by groundedness (Eq. 2) deliberately pushes flagged samples toward zero, sharpening the hallucination decision at some cost to calibration: on this test set the verifier buys hallucination detection and pays for it in calibration. At 175 test samples over 10 bins (≈ 17 –18/bin) bin-level ECE is noisy regardless, so we do not read a strong conclusion either way – exactly the kind of metric disagreement Section 6’s audit is meant to surface, not paper over.

The LoRA-tuned backbone (last row) is a separate generative model (`medgemma-4b-it`), QLoRA fine-tuned for 3 epochs and evaluated by generating and JSON-parsing structured output on the same test split. It is not simulated, reaching 0.707 ± 0.055 accuracy / 0.822 ± 0.041 F1 (mean $\pm$ std, 3 seeds: per-seed 0.766/0.865, 0.697/0.818, 0.657/0.783). On the seed-42 run, per-epoch validation loss was lowest at epoch 1, but epoch 3’s test accuracy came out higher (0.766 vs. 0.720) when both checkpoints were directly evaluated – so we report the final-epoch adapter for every seed rather than quietly picking whichever metric looked better. The spread across seeds (0.657–0.766) is substantial relative to the effect sizes elsewhere in this table, and should be read as a genuine limitation of a single QLoRA fine-tune at this data scale.

8 Error Analysis

Since Med-GReF’s contribution is about evidence and hallucination detection, not classification accuracy alone, we walk through individually-inspected test-set predictions rather than aggregate metrics only. All five cases below are genuine held-out predictions from the Med-GReF (Full) seed-42 model, none constructed for this section.

Correctly grounded (breast-cancer theme): the model passes a genuine sample with high confidence and groundedness, correctly – though the study is in fact a *cardiac* PET/CT scan (also a compound-figure-filter miss, discussed below), with breast-cancer support coming only from a caption-text nodal lesion and carcinoma history, so the “Yes” label reflects rule-based caption matching rather than clinician-validated ground truth.

260 **Correctly-caught hallucination** (breast-cancer theme): the underlying image is a tumour-
261 segmentation panel, while the constructed claim (“cervical cancer is supported,” borrowed from
262 a different donor sample) is paired with spinal-cord-compression MRI evidence text. The verifier
263 assigns low groundedness (0.386) and flags it correctly.

264 **Verifier false positive and a difficult case** both show the same asymmetry. A genuine ovarian-
265 cancer sample the classifier labels correctly is vetoed by the verifier (groundedness 0.558, just under
266 τ) because it is actually a four-panel series of intraoperative *photographs*, not radiology (also a
267 compound-figure-filter miss, see below). Separately, a genuine uterine-fibroid CT the classifier gets
268 right at low confidence (0.230) is flagged at groundedness 0.437. Together the two show the verifier’s
269 error budget on this test set is asymmetric – more willing to reject a genuine sample than to pass a
270 constructed one, the safer direction to err but one that costs accuracy on the final answer.

271 **Verifier false negative** is the most safety-relevant case: a constructed mismatch (donor claim “preterm
272 birth is supported,” with embryo-transfer/gestational-age evidence text, against a true image that is an
273 automated brain-volumetric MRI report “in a woman with polycystic ovary syndrome”) passes with
274 both high confidence and high groundedness (0.998 each) – a high-confidence miss, the failure mode
275 that matters most clinically, since nothing about the model’s own output would prompt a second look.
276 We report it as an open failure case rather than speculate past what the data supports.

277 **Sample outputs.** Each prediction returns a full structured record – retrieved evidence,
278 reasoning/conclusion strings, raw and verifier-calibrated confidence, and a boolean hal-
279 lucination flag – so every answer carries its own audit trail; full records are in
280 `outputs/sample_structured_output.json`.

281 **Compound/multi-panel and low-resolution figures** are excluded at the vetting
282 stage (`data/vetting.py`) before schema construction: of 27,370 candidates, 10,124
283 are caption-flagged as compound figures and 1,337 fall below the 224px floor
284 (`data/processed/vetting_stats.json`). But the cardiac PET/CT and surgical-series
285 cases above both reached the model anyway, since their captions letter panels “A–E”/“A–D” rather
286 than the heuristic’s bracketed lower-case markers – the filter is imprecise in both directions, and
287 widening the marker set is straightforward future work not retrofitted into the numbers reported here.

288 9 Limitations and Broader Impact

289 The test set stays modest in size (175 samples), and coverage across the fifteen taxonomy conditions
290 is uneven even after the expansion in Section 3 (twelve have test-split representation, two have a
291 handful of samples but no test coverage, one has none at all); we scope every claim to conditions with
292 actual test-split representation and name the excluded ones explicitly. Question/evidence/reasoning
293 fields are rule-derived from captions, not clinician-authored – a clinician spot-check of a held-out gold
294 subset is planned but not yet done, and no claim here should be read as clinically validated. Seed-level
295 SD (Table 1) should be read alongside the bootstrap resampling-variance estimate in Section 7;
296 external validation on a second, independent dataset was requested but hasn’t been attempted, since
297 the only larger corpus known to the authors was not available to fetch. **This system is a research
298 prototype and is not validated for, and must not be used for, real clinical decision-making;** its
299 intended use is as an evaluation and error-analysis tool for studying hallucination in medical VLMs,
300 not a diagnostic aid. The underlying corpus comes from publicly available, de-identified PubMed
301 Central figures under open-access licenses, so our pipeline introduces no patient-identifiable data.

302 10 Conclusion

303 Med-GReF combines evidence-guided multimodal fusion with a dedicated hallucination verifier
304 on an actively-audited PubMed Central women’s-health imaging corpus, delivering a measurable
305 accuracy gain (accuracy $0.817 \pm 0.006 \rightarrow 0.912 \pm 0.020$, ROC-AUC $0.570 \pm 0.080 \rightarrow 0.881 \pm 0.016$)
306 alongside a substantial cut in undetected hallucinations ($2.4\times$) at negligible extra accuracy cost. We
307 treat the audit process that closed an earlier accuracy/robustness trade-off (Section 7) as part of the
308 contribution itself, since medical-AI results are only as trustworthy as the evaluation code behind
309 them.

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A Technical Appendix

Additional per-condition dataset statistics, the full fetch/vetting pipeline configuration, and code for reproducing every number in Table 1 are available in the accompanying repository (link withheld for anonymized review; to be added at camera-ready).

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